

ABSTRACT

Large Language Models (LLMs) have strong sequence-modeling capability, but their use in time-series forecasting requires evaluation beyond average error metrics. This work builds on the Time-LLM framework, which adapts frozen pre-trained LLMs to numerical forecasting through patch embedding, reprogramming, prompt-as-prefix conditioning, and output projection.

We study whether BTC forecasting remains stable when historical training windows and sequence lengths change, and whether internal TimeLLM signals can identify safer forecasts through risk-aware selective forecasting. Results show partially robust directional accuracy, non-monotonic growing-window behavior, stronger compact-context performance, and useful hidden-state plus time-series signals for forecast-level reliability ranking.

MOTIVATION

- Average forecasting metrics can hide row-level reliability variation.
- Cryptocurrency markets are non-stationary; older context may help, hurt, or become stale.
- Training-window choice, sequence length, and forecast uncertainty should be studied together.
- A useful forecasting system should predict values and identify forecasts that are more reliable.
- The key question is whether TimeLLM-style forecasts are robust and risk-aware.

TIME-LLM FORECASTING MODEL

The Time-LLM framework bridges numerical time series and frozen LLMs without full model retraining.

- Input embedding:** normalize values and divide inputs into patches.
- Reprogramming:** map numerical patches into the LLM token space.
- Prompt-as-prefix:** add task context, instructions, and statistics.
- Frozen backbone:** process the sequence using a pre-trained LLM.
- Output projection:** convert LLM representations into forecasts.

Extension: this poster evaluates temporal robustness, sequence-length sensitivity, and risk-aware forecast selection.

KEY FINDINGS

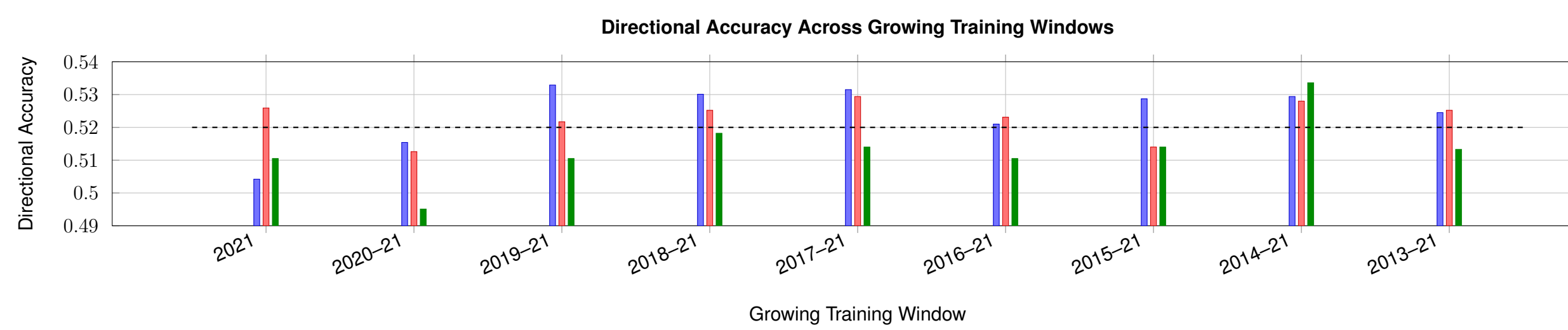
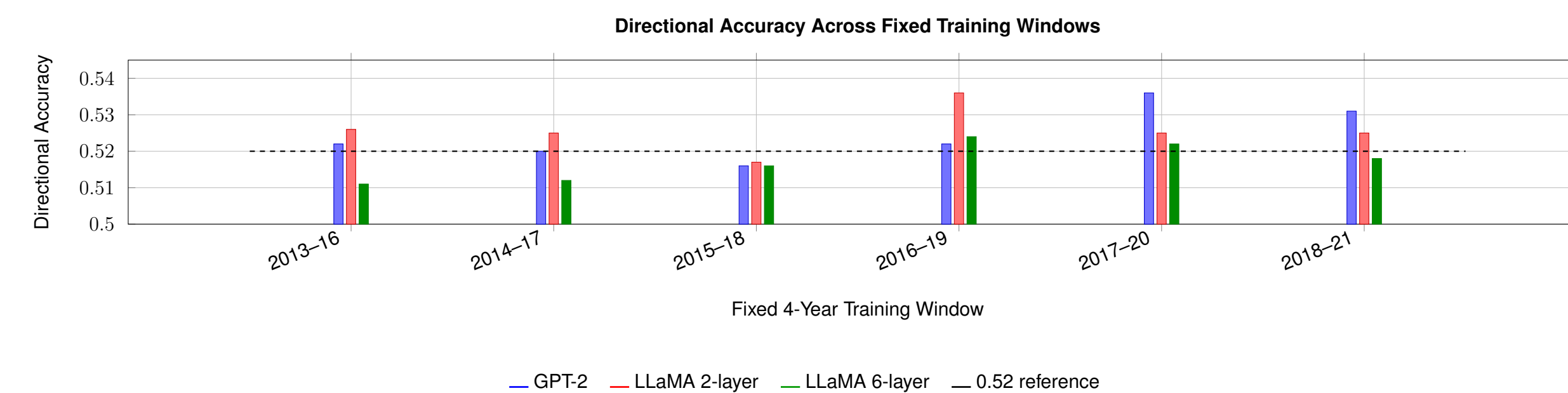
- BTC directional accuracy was **partially robust** across fixed and growing training windows.
- More historical data did **not** consistently improve forecasting performance.
- The best growing-window result came from LLaMA 2-layer using only the most recent training year.
- Lower sequence-length ablations showed compact recent context can outperform longer history.
- Internal TimeLLM signals helped rank forecast reliability.
- Combined time-series and hidden-state features were strongest for direction-based selection.

EXPERIMENTAL SETUP

- Dataset:** Kraken BTC/USD daily OHLCV.
- Forecast target:** future close price.
- Fixed test window:** 2022-01-01 to 2025-12-31.
- Training windows:** fixed 4-year windows and growing windows ending in 2021.
- Backbones:** GPT-2 and LLaMA3.1 configurations.
- Metrics:** MAE, RMSE, and Directional Accuracy.
- Risk-aware evaluation:** MLP and XGBoost selectors evaluated at 90% coverage.

TEMPORAL ROBUSTNESS RESULTS

The BTC study keeps the same future test period and varies the historical training window. Fixed-window and growing-window tests both show partially robust directional accuracy, while error metrics are more sensitive to history length and recency.



Interpretation: fixed-window and growing-window results both show **partially robust** directional accuracy. Values fluctuate across eras, but most remain near 0.52–0.53, suggesting directional prediction is more stable than MAE/RMSE.

SEQUENCE-LENGTH ABLATION

Follow-up sequence-length studies showed that reducing input context can improve performance. These are related ablations, not one identical experiment family.

Experiment Family	Best Error	Best DirAcc	Main Finding
Early sweep	seq_len=14	seq_len=28	shorter helped
Lower range, pred_len=8	seq_len=11	seq_len=11	best lower run
Short horizon, pred_len=4	seq_len=4	seq_len=6	compact context best

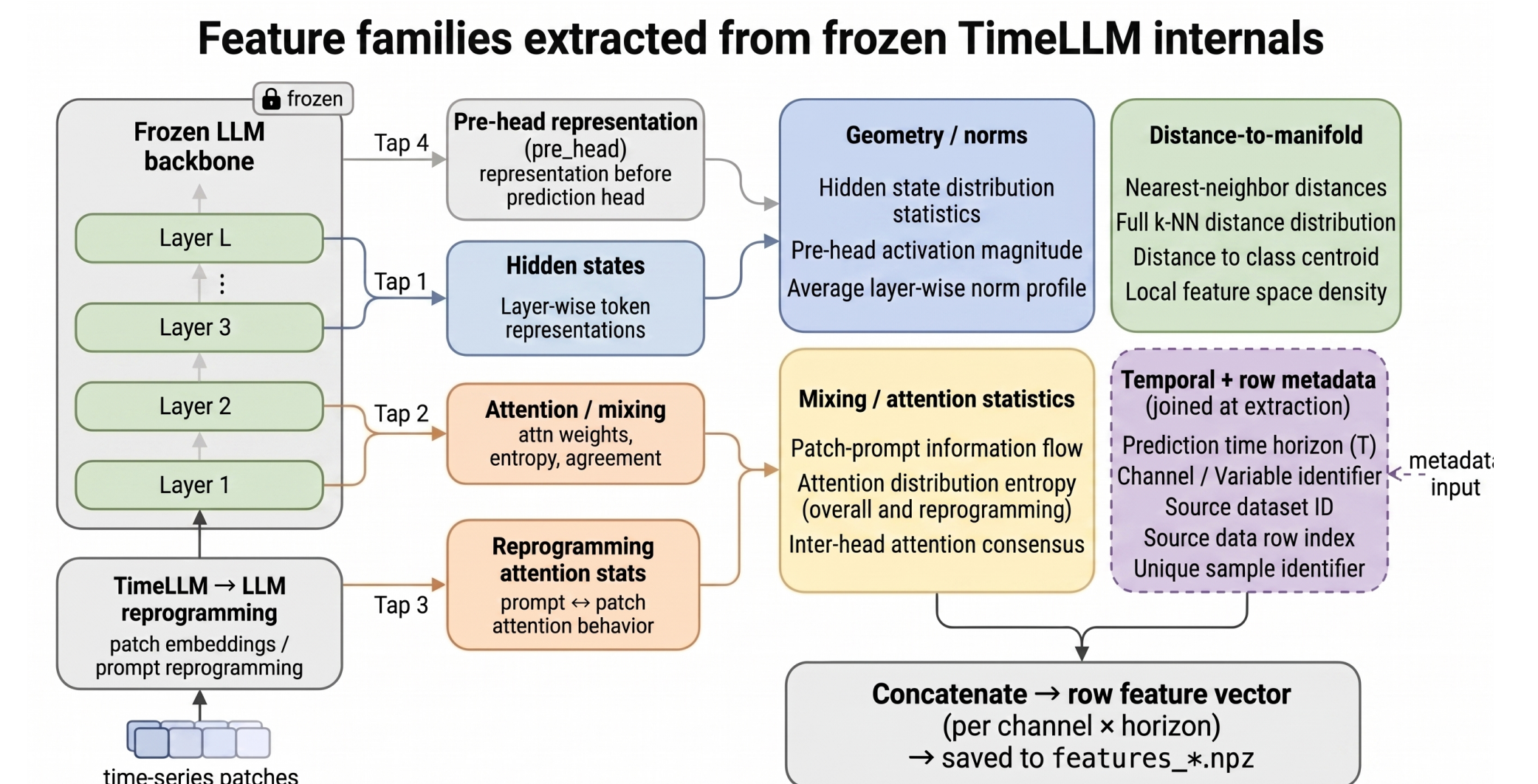
Seq.	MAE	RMSE	DirAcc
13	2234.63	3250.56	0.5252
12	1923.92	2754.68	0.5311
11	1726.44	2491.58	0.5349
10	1938.97	2806.47	0.5331
9	1795.22	2598.94	0.5231
8	1797.86	2613.42	0.5296

Seq.	MAE	RMSE	DirAcc
4	1409.68	2086.25	0.5240
5	1508.60	2218.99	0.5326
6	1630.60	2385.97	0.5420
7	1718.16	2504.06	0.5410

Takeaway: compact recent context can outperform longer input history. In the standard lower-range family, seq_len=11 was strongest; in the short-horizon variant, seq_len=4 minimized error and seq_len=6 maximized directional accuracy.

RISK-AWARE SELECTIVE FORECASTING RESULTS

Risk-aware selective forecasting treats each forecast row as a reliability decision. The selector ranks forecasts from safer to riskier and retains the top fraction at fixed coverage.



Target	Best Selector	Relative Gain
ETTh1 MDA	Combined + XGBoost	3.18%
ETTh2 MDA	Combined + XGBoost	3.41%
ETTh1 MAE	Hidden + MLP	13.46%
ETTh2 MAE	Combined + XGBoost	6.42%

- Feature arms:** time-series metadata/output signals, hidden-state TimeLLM representations, and combined fused features.
- Selector impact:** MLP and XGBoost improved retained-set quality at 90% coverage; combined features were strongest for direction selection.

CONCLUSION

LLM-based time-series forecasting should be evaluated not only by average accuracy, but also by **robustness over time** and **forecast-level reliability**. BTC results show that more history is not automatically better: growing-window and sequence-length studies suggest recent compact context can be more useful than long historical context in non-stationary crypto markets.

FUTURE WORK

- Extend BTC experiments to ETH and other assets with broader walk-forward validation across market regimes.
- Apply risk-aware selectors directly to crypto forecasts and compare with conformal prediction and ensemble uncertainty methods.

REFERENCES

- tiny [1] Jin et al. *Time-LLM*, ICLR 2024.
[2] Temporal robustness and BTC forecasting, 2026.
[3] Risk-aware selective forecasting from TimeLLM hidden states, 2026.
[4] Geifman and El-Yaniv, NeurIPS 2017.